

SMART VIT HACKATHON 2026



- **Problem Statement ID – SVH26008**
- **Problem Statement Title- Temple and Pilgrimage**

Crowd Management

- **Theme- Heritage and Culture**
- **PS Category- Software**
- **Team ID- svh-10125**
- **Team Name - Nexus Six**





TeerthMitra is a unified, AI-powered platform for real-time crowd monitoring, predictive safety, and pilgrim guidance across temple towns — built for Somnath, Dwarka, Ambaji, and Pavagadh, and designed to scale to any pilgrimage site in India

MONITOR

- **Digital Venue Map / Digital Twin** — gates, chokepoints, queues, parking zones, with an accessibility overlay for elderly & differently-abled routes
- **Crowd Monitoring Module** — live density & flow per zone
- **Digital Queue & Ticketing (Group Host Booking)**— bookings feeding live occupancy data
- **Digital Tether** — real-time group/family positions via Bluetooth; QR wristbands track children and elderly pilgrims, linked to guardian's phone

FORECAST

- **Crush-risk Prediction** — density + flow readings converted into a per-zone risk score, weighted higher at known chokepoints
- Compares against historical festival-day patterns to project congestion 15–30 minutes ahead, flagging Safe / Caution / Critical
- **Admin/Control Dashboard** — simulates the impact of options (e.g., gate closure, reroute) before recommending one

RESPOND

- **Alert & Notification System** — pushes warnings to marshals
- **Emergency Response Routing** — dispatches nearest medical/security team on SOS
- **Pilgrim App/Portal** — live wait times, safer routes, less-crowded slots
- **Digital Tether** — triggers Reunion Mode for separated groups; QR-wristband scan instantly reunites lost children/elderly with guardians
- **Dynamic Smart Parking & Shuttle Tracking** — reroutes incoming traffic before roads jam

TARGETED IMPACT

- **Eliminates Overcrowding**
- **Proactive Stampede Prevention**
- **De-congests Transit**
- **Zero Lost Pilgrims**

CORE INNOVATIONS

- **Predictive Safety:** Foresees crowd surges before choke points become fatal.
- **Offline-First Connectivity:** Bluetooth-based tracking circumvents congested cellular networks.
- **Digital Tether & QR Wristbands:** Failsafe, low-cost reunion mechanism for children and elders.
- **High-Precision Lane Navigation:** Bypasses GPS limitations in dense, covered shrine pathways.





TECHNICAL APPROACH

TeerthMitra

TECHNICAL DIFFERENTIATORS

- **Proactive vs. Reactive Safety:** Predicts stampede/crush risks 15–30 minutes ahead, allowing authorities to act before critical density is reached.
- **Offline-First Connectivity:** Bluetooth RSSI micro-tethering works inside narrow stone corridors and survives cellular network crashes during peak festival surges.
- **Inclusive Low-Cost Safety Net:** QR wristbands protect non-smartphone users (children and elderly) at negligible cost without expensive wearable hardware.

MULTI-SOURCE INGESTION PIPELINE

- Visual Telemetry (CCTV)
- Digital Ticketing & Access Logs
- Transit & Parking Sensors

SPATIAL GIS & DIGITAL TWIN SYNTHESIS

- Digital Twin
- Accessibility Overlays

CELLULAR-INDEPENDENT MICRO-LOCATION(DIGITAL TETHER)

- Zero-GPS Dependency
- QR wristband Registry

CROWD DENSITY & VELOCITY VECTOR ENGINE

- Calculates instantaneous spatial density
- Computes velocity vectors across high-density choke points using optical flow? spatial frame analysis

15-30 Minute Early Crush-Risk Prediction Engine

- Time-series ML Models
- Tri-color Risk Indexing
green : safe
yellow : caution
red : critical
- Runs graph-based flow simulations

COMMAND & EMERGENCY DISPATCH

- Marshal Push Alert
- Emergency SOS Routing

PILGRIM EXPERIENCE & TRANSIT BALANCING

- Pilgrim App/Portal
- Dynamic Smart Parking & Transit Rerouting

ZERO-NETWORK REUNION MODE

- Offline Directional tethering
- QR Reunion System

TECH-STACK

FRONTEND:
Flutter, Dart, MAterial UI

BACKEND:
FastAPI, Python, Rest API

DATABASE:
PostgreSQL, Firebase Firestore

AI/ML:
YOLOv11, OpenCV, TensorFlow, Scikit-Learn

CLOUD AND ANALYTICS:
AWS, Firebase FCM, Power BI, Grafana

IoT & HARDWARE:
ESP32, QR CODE



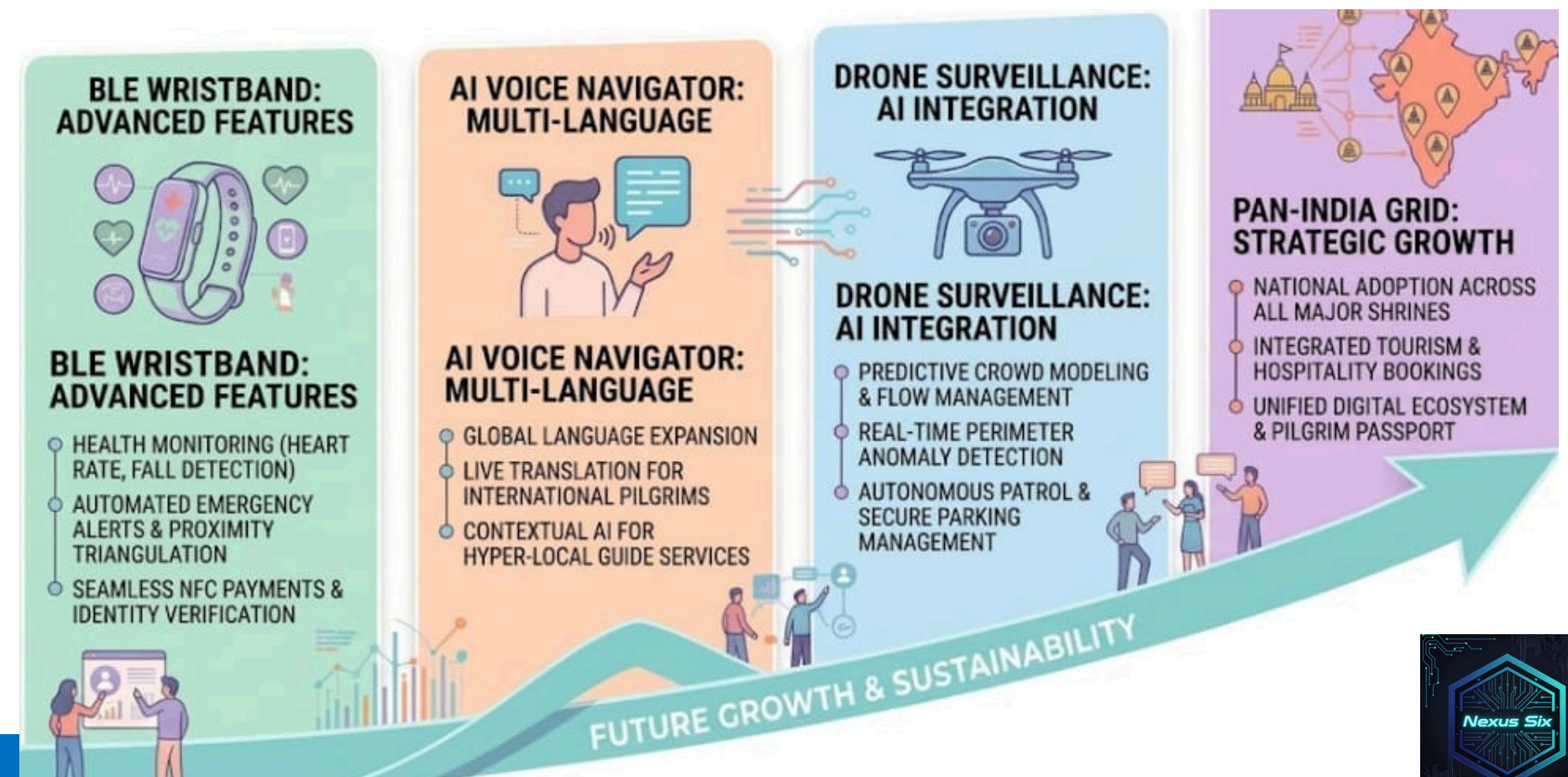
FEASIBILITY

- **Utilises existing CCTV infrastructure** - no new cameras needed at pilot sites
- **Offline first design** - Digital Tether and QR wristbands work without cellular data, directly solving the network overload problem festivals cause
- **High-precision lane navigation** — Bluetooth proximity bypasses GPS's known failure in narrow, covered shrine pathways, where satellite signal is unreliable
- **Modular architecture** — Monitor, Forecast, and Respond modules can be built and deployed independently for staged rollout
- **Scalable, subscription-style licensing** model per temple site is a realistic long-term revenue path
- Traffic-light dashboard (Safe/Caution/Critical) needs **no specialized training** for marshals or volunteers

RISKS & MITIGATION

- **Network Overload:** Digital Tether and QR wristbands function completely offline.
- **GPS Dead Zones:** Uses Bluetooth proximity instead of GPS for narrow, covered lanes.
- **Massive Scale:** Powered by a horizontally scalable, cloud-native AWS backend.
- **Damaged Bands:** Wristbands store no data directly and are cheap to re-issue instantly.
- **Power Failures:** Kiosks run on battery backup while logic computes in the cloud.
- **False Alarms:** Requires sustained crowd density anomalies to prevent false alerts.
- **Digital Literacy:** Pilgrims need no app or tech—staff handle all scanning operations.
- **Privacy Concerns:** Uses anonymized density counting with encrypted guardian data.
- **Camera Blind Spots:** Digital Twin flags unmonitored zones to alert ground staff.

FUTURE SCOPE



BENEFITS :

Economic Impact

- Lower operational costs
- Increased local trade
- New revenue streams through optimized crowd flow

Social Impact

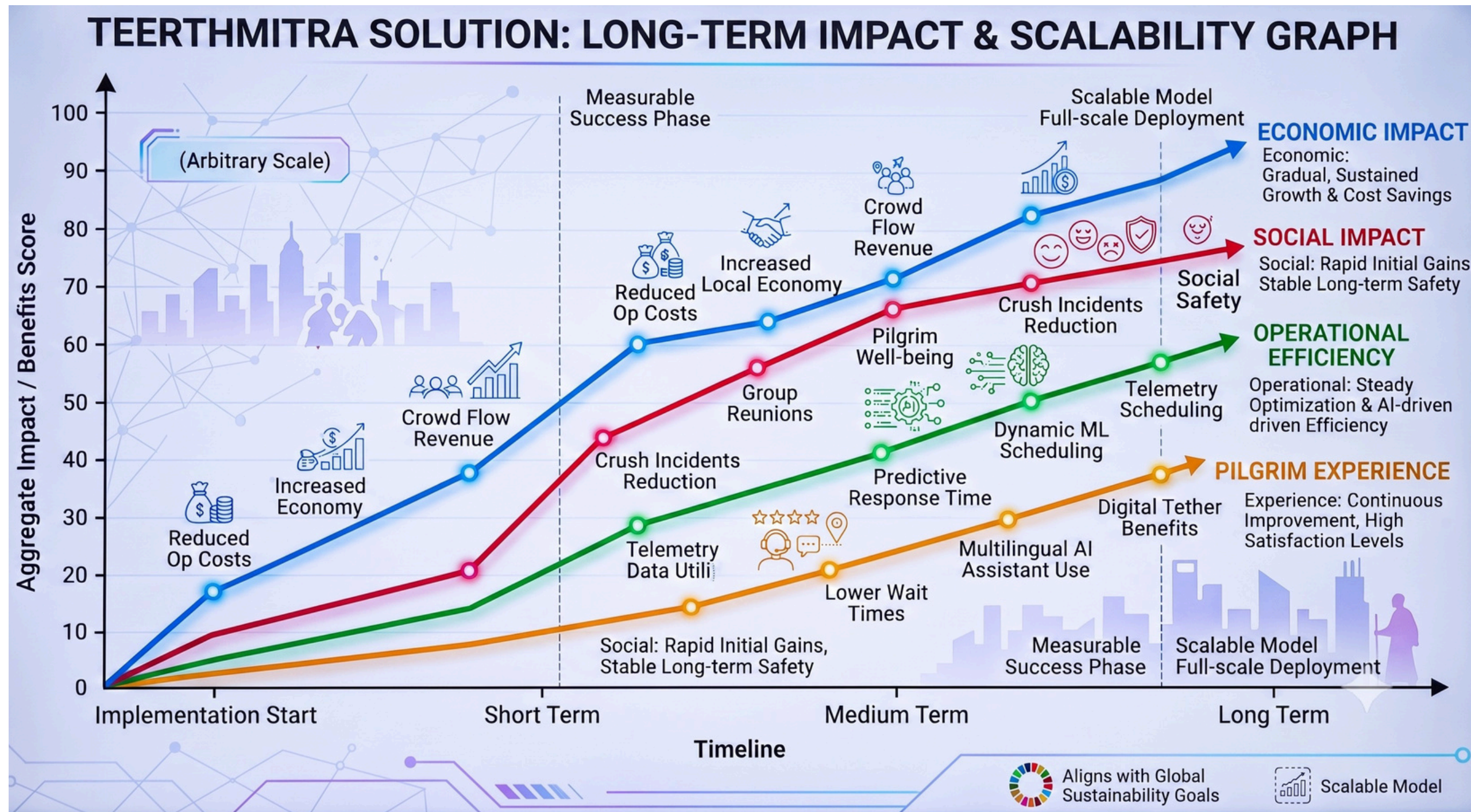
- Major reduction in crush/stampede risks
- Faster group reunions
- Enhanced overall pilgrim safety

Operational Efficiency

- AI-driven ML scheduling
- Real-time telemetry data
- Predictive response times for smoother management

Pilgrim Experience

- Shorter wait times
- Multilingual AI assistance
- Digital tethering for continuous satisfaction



GOOGLE DRIVE LINK : <https://drive.google.com/drive/folders/1Xr3NQnyYzKyadMcKMbZn5zWadM66Tcdn>

- **RESEARCH PAPER ON TEMPLE AND PILGRIMAGE CROWD MANAGEMENT :**

Link : <https://jsrtjournal.com/index.php/JSRT/article/view/462>

<https://www.ijraset.com/best-journal/temple-and-pilgrimage-crowd-management>

Learning : Combining AI-based computer vision with IoT sensors enables real-time crowd density tracking and automated time-slot tokening to prevent overcrowding and stampedes.

- **SMART CROWD MANAGEMENT FRAMEWORK :**

Link : <https://www.triptotemples.com/news/announcements/kpmg-queues-on-crowd-management-solutions-for-indian-pilgrimage-sites>

Learning : Modernizing Indian pilgrimage crowd management requires combining digital slot-booking

- **PREDICT CRUSH RISK BEFORE IT BECOMES A STAMPEDE :**

Link : <https://www.mcrhrdi.gov.in/itpbatch2/week2/PILGRIM%20STAMPEDES%20AND%20CROWD%20CONTROL.pdf>

Learning : Effective pilgrim crowd control requires managing structural bottlenecks, preventing panic triggers, and monitoring crowd density to avoid fatal upright asphyxiation

- **DIGITAL ETHER**

Link : [https://www.researchgate.net/publication/396705094_Evaluation_of_RSSI-](https://www.researchgate.net/publication/396705094_Evaluation_of_RSSI-Based_Distance_Estimation_with_ESP32_BLE_Modules_for_Indoor_Asset_Tracking)

[Based_Distance_Estimation_with_ESP32_BLE_Modules_for_Indoor_Asset_Tracking](https://www.researchgate.net/publication/396705094_Evaluation_of_RSSI-Based_Distance_Estimation_with_ESP32_BLE_Modules_for_Indoor_Asset_Tracking)

Learning : Safe temple crowd management relies on strict venue capacity thresholds paired with segregated, one-way entry and exit routes to prevent dangerous crowd stagnation and compression.

- **CAPACITY PLANNING & FLOW SEGREGATIONI**

Link :: <https://ndma.gov.in/sites/default/files/PDF/Reports/managingcrowdsguide.pdf>

Learning : Safe temple crowd management relies on strict venue capacity thresholds paired with segregated, one-way entry and exit routes to prevent dangerous crowd stagnation and compression.